SUSTAINABILITY



COOLING ENVIRONMENTS



Building a Scalable Waste Heat Ecosystem

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Where this session goes, and what you leave with?

01	**WHR is evolving** Permitting requirements and an Offtaker Rolodex beyond district heating
02	**Defining a possible decision-making framework** Inputs, go / no-go gates, weighted scoring
03	**Optimizing offtakers for local communities** Examples of optimization with weighted scoring
04	**Matchmaking needs to be standardized** A marketplace with common language and shared data
05	**What's next** What else needs to be standardized and how you can help

You leave with an example of how structured gates and scoring can be used to evaluate heat reuse opportunities.
What offtakers can provide communities, which gates stop a project, how value is weighted regionally, and how to connect heat supply and demand.

Waste Heat Recovery Is Becoming a Permit Requirement



EU + GERMANY | BINDING

Reporting has become quantified reuse targets

- EU rules require waste-heat reporting and installation-level cost-benefit analysis.
- Germany raises its Energy Reuse Factor target from 10% in 2026 to 20% in 2028 for data centers above 200 kW.



PERMITS + VIRGINIA | EMERGING

WHR now affects permitting and market access

- Permit related delays could mean revenue loss of millions of dollars to DC operators
- Virginia HB2578 signaled growing U.S. policy interest, although the 2025 bill failed and is not in force.

The Field Is Moving Beyond District Heating

TODAY | ESTABLISHED

District heating dominates the model

- Most deployed and studied projects route waste heat into municipal or campus district heating networks.
- It works where dense networks, cold climates, and willing utility partners already exist.
- That leaves large parts of our fleet without a viable local offtaker.

EMERGING | EXTERNAL SOURCES

Alternatives are being actively mapped

- OCP's Waste Heat Reuse 101 catalogs offtakers well beyond the district heating case.
- "Flipping the switch" frames waste heat as a route to carbon-negative and water-positive data centers.
- Independent analyses converge on a common shortlist of promising non-district offtakers.

A Waste Heat Recovery Options beyond District Heating

Per 1 MW-yr...

DISTRICT HEAT

District heating

Established, but limited by network proximity and utility readiness.

~650 homes heated

INDUSTRIAL HEAT

Industrial process heat

Compatible demand nearby, with a willing owner, is the fastest offtake.

8.76 GWh delivered

WATER PATHWAYS

Water treatment & generation

Supports water resilience where source water or mature tech exists.

255,000 – 365,000 m^3 water treated

6,000 – 13,500 m^3 water generated

CARBON REMOVAL

Direct air capture

Low-grade heat improves DAC economics where a CO_2 pathway exists.

2,000–5,000 t CO_2 captured

AGRICULTURE

Greenhouses & aquaculture

Enables year-round growing where land and an operator exist.

~2 hectares of greenhouses

Standardized workflows can turn options into defensible choices

A workflow option: gate feasibility first, then score to optimize the decision.

1 INPUTS

- Heat supply/demand requirements
- Local community needs
- Market
- Land availability/Proximity
- Regional policy

2 GATE

- Technical fit
- Operational boundary
- Market readiness
- Regulatory

3 SCORE

- Economics
- Carbon
- Water
- Community

4 DECIDE

- Include or reserve space for WHR in datacenter design
- Go/No-Go Decisions

Gates are pass / fail. Only options that clear every gate are scored, so the final decision is always defensible.

Excepted benefits can only be realized if an offtaker is deployable

2

GATE

TECHNICAL



Can available temperature, flow and duty cycle serve the load?

GO / NO-GO

OPERATIONAL



Can the interface remain isolated, fail-safe and independent per offtaker?

GO / NO-GO

MARKET



Is there a real owner, operator, customer and route to market? Land/Proximity?

GO / NO-GO

REGULATORY



Does the pathway align with local requirements and approvals?

GO / NO-GO

Only eligible pathways enter value optimization.

Once we know what's possible, We can optimize

3 SCORE

Scoring gives eligible pathways a common language for comparison

$$S_{Total} = (w_{econ}S_{econ}) + (w_{carbon}S_{carbon}) + (w_{water}S_{water}) + (w_{social}S_{social})$$

ECONOMIC

ROI and payback from the datacenter perspective

CARBON

Carbon capture and emissions avoided, minus the project's embodied and operational carbon footprint

WATER

Water generated and saved, minus the project's embodied and operational water footprint

SOCIAL

Community benefits, including jobs and equity of access to essential resources

w = weight based on company priorities , S = Score for specific offtaker and location

DEVELOPED WITH VILLANOVA UNIVERSITY ES2 Center for Energy-Smart Electronic Systems

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Location Affects the Subscores

3 SCORE

$$S_{Total} = w_{econ}S_{econ} + w_{carbon}S_{carbon} + w_{water}S_{water} + w_{social}S_{social}$$

Location 1

	WEIGHT	FOOD & BEVERAGE SUBSCORE	WATER GENERATION SUBSCORE
Economic	0.25		
Carbon	0.25		
Water	0.25		
Social	0.25		
Total			

Location 2

WEIGHT	FOOD & BEVERAGE SUBSCORE	WATER GENERATION SUBSCORE
0.25		
0.25		
0.25		
0.25		

Location Affects the Subscores

3 SCORE

$$S_{Total} = w_{econ}S_{econ} + w_{carbon}S_{carbon} + w_{water}S_{water} + w_{social}S_{social}$$

Location 1

	WEIGHT	FOOD & BEVERAGE SUBSCORE	WATER GENERATION SUBSCORE
Economic	0.25	80	45
Carbon	0.25	85	40
Water	0.25	70	35
Social	0.25	55	60
Total			

Location 2

WEIGHT	FOOD & BEVERAGE SUBSCORE	WATER GENERATION SUBSCORE
0.25	45	85
0.25	35	75
0.25	40	70
0.25	55	60

Location Affects the Subscores

3 SCORE

$$S_{Total} = w_{econ}S_{econ} + w_{carbon}S_{carbon} + w_{water}S_{water} + w_{social}S_{social}$$

Location 1

	WEIGHT	FOOD & BEVERAGE SUBSCORE	WATER GENERATION SUBSCORE
Economic	0.25	80	45
Carbon	0.25	85	40
Water	0.25	70	35
Social	0.25	55	60
Total		72.5	45

Food & Beverage wins when partners are already located nearby.

Location 2

WEIGHT	FOOD & BEVERAGE SUBSCORE	WATER GENERATION SUBSCORE
0.25	45	85
0.25	35	75
0.25	40	70
0.25	55	60
	43.8	72.5

Atmospheric water generation wins when land is available for co-location.

The same framework produces different regional answers

SCORE 3

$$S_{Total} = w_{econ}S_{econ} + w_{carbon}S_{carbon} + w_{water}S_{water} + w_{social}S_{social}$$



URBAN HEAT NETWORK

Dense, cool-climate city

BEST-FIT PATHWAYS

District heat + Industrial reuse



WATER-STRESSED REGION

Scarce water; compatible feed or climate

BEST-FIT PATHWAYS

Water treatment + Atmospheric Water Generation



LAND + MARKET ACCESS

Operator, land and year-round demand

BEST-FIT PATHWAYS

Greenhouses + Industrial reuse

A repeatable framework for evaluating waste heat opportunities

One possible sequence for evaluating and prioritizing opportunities

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- Local community needs
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4 DECIDE

- Include or reserve space for WHR in datacenter design
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This process is only useful if it turns into results

A marketplace needs common language, contracts, and shared data

COMMON LANGUAGE

Shared definitions for temperature grades, capacity and availability

Clear options for demarcation points: where supplier ownership ends and offtaker ownership begins

STANDARD CONTRACTS

Template heat purchase agreements, tariffs and risk allocation

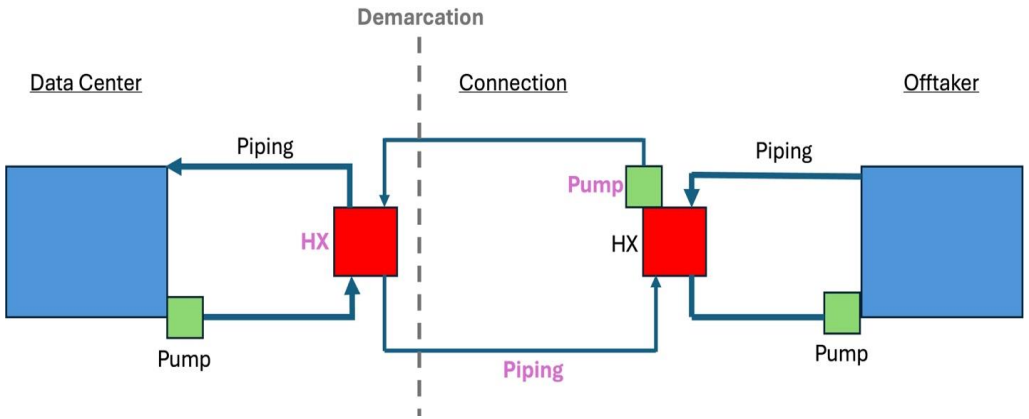
Repeatable terms so each deal is not negotiated from zero

SHARED DATA

Offtaker demand data is highly local and slow to collect

Pooling it into a common marketplace turns years of site work into a lookup

EXAMPLE | Demarcation of supplier vs. offtaker ownership



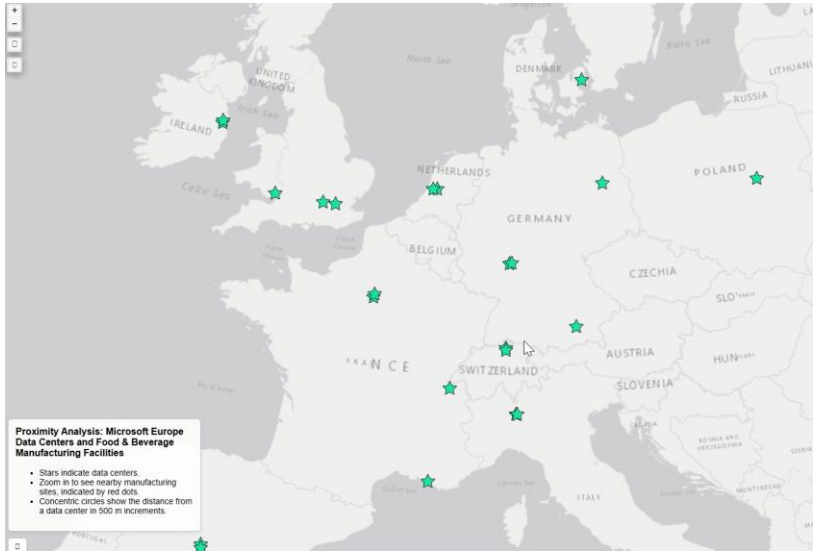
THIS IS ALREADY UNDERWAY

OCP, NZIH, and other industry groups are building WHR marketplace frameworks today

The opportunity is to converge on one set of definitions and data schemas rather than fragment across them

We can achieve this for one location. What if we scaled?

DEMAND · F&B PLANTS NEAR OUR SITES



READINESS · CO_2 NETWORK



WHAT THAT DE-RISKS

100s MW potentially unlocked
IN GERMANY

1,000s GWh/yr HEAT RETURNED

~50,000 homes OFF GAS HEAT

Subject to assumptions

This is the resolution a matchmaking platform needs. One country took months. Every site needs it, and it has to live under one common umbrella.

What could the broader OCP community standardize?

Heat-source descriptor

Temperature, capacity, availability, duty cycle, interface

Demand descriptor

Temperature need, load profile, proximity, readiness

Feasibility gates

Technical, operational, regulatory, commercial

Value metric scores

Carbon, water, economics, community

Evidence governance

Source, confidence, validation owner, assumptions

Marketplace data structure

Interoperable data structure for matching

Standardized contracts

Template offtake terms, risk allocation, pricing structure

OPEN • TRANSPARENT • REPEATABLE • REGIONALLY ADAPTABLE

Move from isolated projects to a scalable waste heat ecosystem

CONNECT ON LINKEDIN
Maria Viitaniemi • Bharath Ramakrishnan

Three ways the OCP community can turn this framework into shared practice.

SHARE DATA

- Regional use cases
- Supply + demand datasets
- Metric definitions

TEST AND REFINE THE FRAMEWORK

- Apply gates + scoring
- Compare regions
- Expose edge cases

HELP DEFINE STANDARDS

- Common descriptors
- Evidence conventions
- Marketplace fields



Join the OCP Waste Heat Reuse sub-group

Open to all OCP members — contribute data, test the gates, and help set the standard.

Thank You!

